

Review of NoSQL Data Stores: Using a reactive three-tier application for software developers to achieve a high availability application design architecture

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Abstract— Every software developer needs to understand that the application architecture is essential. Despite this, many software developers follow the design of software architects and believe that software architects are always right. Changes in the design and application requirements have changed in the last years. However, with the rise of the World Wide Web (WWW), software developers had to store more data on servers. Relational databases cannot scale the data fast enough on web servers. Applications are deployed on every mobile device to cloud devices hosted on servers, and because of this change, users expect faster response times. There is minimal empirical evidence to help developers choose which data store to use. In addition, more NoSQL data stores are added to the ecosystem. In this paper, we used a systematic review to collect evidence on the best performing data store for a highly available application design architecture. This review allowed us to study experiments performed by researchers on NoSQL data stores namely wide-column, key-value, graph and document stores and which programming language driver to use for implementing such research. Our empirical findings suggest to develop a conceptual framework to design a reactive three-tier application for software developers to achieve a high availability application design architecture.

Keywords— *NoSQL, reactive, applications, databases, developers, data, three-tier applications, cloud*

I. INTRODUCTION

NoSQL data stores are currently the web's most used data stores [4], because these data stores are known to have a few key facts, such as high availability against failures, and can scale and have high performance systems and highly available [5]. They use replication by storing the data at multiple nodes in a data center or multiple data centers. If a node fails where data reside on, the NoSQL data store can still function using its replicated node(s). Where the data resides this means the system can handle fault tolerance and can be classified as highly available [4].

This research focused with the help of existing literature to help us identify the gap in the existing empirical knowledge of the current study. The study observed NoSQL

distributed systems running on commodity servers using distributed consensus on each data store [5].

The study described the most relevant discussions surrounding this research study and brought the most important aspects to light in order to point out the current work that was done which directly affected our study.

There are many data store platforms on NoSQL data stores. Each of these data stores use their own architecture and data storing scheme and how to implement. Another article about the reliability and availability properties of distributed database systems based on database design options. *"Reference [8] Reliability and availability of database systems have also been considered by other authors. In this document, however, we are facing the problem that publications on that topic are either pure marketing material or on the other hand scientific publications addressing only a single aspect of the entire story"*.

Not enough literature about how non-relational databases are and how each database functions, are different from each other based on the developers need, this affects the tools and choosing the right data store [6].

The Not Only SQL (NoSQL) is an alternative, but not to replace relational databases [1]. Thus, the NoSQL and relational databases complement each in database activities and management of data sets [2]. Based on its premises, which include the ability to handle data in a very fast mode using no fixed schemas and unstructured data that are based on meta data to achieve a high performance rate, the NoSQL became more popular [3].

The study described the background relating to the need to use NoSQL and the reason why there's move towards the use of NoSQL data stores. The study focused on the keywords for this study, which include NoSQL, SQL and data sets. In addition, also gave a brief overview of this study relating to

NoSQL data stores and the relational data stores through a review of existing literature.

The discussion included a review of the literature. Which found that NoSQL data types could read and write effectively. To support applications that required high performance systems to be reliable to store the data on more than one computer node in a cluster.

Developers have to select a chosen database based on the requirements of the data structures. A lot of time could be saved if known which database to use. [6] The absence of availability of an analysis framework and the actual empirical data will continue to pose a challenge on the uptake of appropriate NoSQL databases. Users will continue to be subjected to biased analysis. [1] states: "there is a large amount of NoSQL candidates, and most have not been compared thoroughly yet." Today there are so many NoSQL data stores and it is very challenging for developers to pick which one to use.

The highly available and reliable NoSQL database per data type category can determine how each NoSQL database performs on inexpensive commodity hardware. Developers use different programming languages with different data types. This research allows to select which NoSQL database to choose and programming languages.

I. Research Methodology

To begin the first step of the systematic review the study identified the goals of this review. The study wanted to find solutions done by other researchers and how they differ from each other and look at the implications they would have on the research study.

In the next phase of the systematic literature review the study had to conduct the review by identifying the research by means of collecting many relevant articles to answer the research questions, the study needed to document each step of the research process to select the primary studies and apply filtering based on the abstract and titles of the articles and research papers.

A quality assessment of these papers was done for the synthesis and analysis, the study then continued the data extraction process where the study collected data from each research paper using a data collection form, which were designed during the development of the review protocol, the goal was to collect all data from papers to answer our research questions.

The last step of our conducting phase was to synthesize the data collected to answer the research questions, the data collected was homogenous and quantitative analysis was performed.

The final phase of the systematic literature review was to specify the dissemination strategy, based on [9], The study had to create a technical report from the systematic review to be made available to the community through a conference or journal and the report had to be evaluated by an expert in the field to be submitted in a paper to be peer-reviewed.

Based on the work done, the presented a systematic review to look at NoSQL database systems on commodity hardware. Normal relational data stores have challenges scaling horizontally. The study needed to look at the problems NoSQL databases solve that relational databases can't solve. There are also challenges presented by NoSQL data stores as there are a vast amount of NoSQL data store systems and the functions of each database differs from each other [10]. This makes it hard for a software developer to choose which data structure to select.

A. Review Method

Based on the objective of the study, to assess the NoSQL from availability and reliability perspectives, documentation technique [11] was employed through which existing literature was collected as data. The systematic review approach was selected for the analysis. The technique and approach were most suitable for the study especially the fact that most research methods and frameworks do exist in the field of computing but software engineers are not exposed to these procedures and frameworks [12]. A systematic review was decided as the research method for this research, as we tried to understand the problem and find or propose a solution on how to solve the problem [13]. The systematic review improved the position of computer science and software engineering in many ways as authors and students benefit having clear procedure-driven research when reviewing the background material for their thesis. And to identify where the background supports or conflicts with the thesis, also to improve the quality of papers and producing a better quality review process [14].

This a repeatable research method and could be replicated by other researchers in the field.

The systematic literature review process was performed in eight steps.

The systematic literature review process

- ST1 Defining research questions
- ST2 Defining the systematic literature review protocol
- ST3 Search for relevant studies
- ST4 Selection of studies
- ST5 Quality assessment
- ST6 Data collection
- ST7 Data synthesis and analysis
- ST8 Dissemination

B. Defining Research Questions

No systematic review has been used for NoSQL databases and data structures of the programming languages used on commodity hardware. Therefore, the study wanted to make a contribution to the research community and developers using NoSQL databases and those that still want to adapt to NoSQL databases for storage. The study decided to identify the existing methods, constraints and approaches

To begin our first step of our systematic literature review we identified our goals of this review.

The study wanted to find solutions done by other

researchers and how they differ from each other. And to look at the implications they would have on the research study:

RQ1: What empirical evidence exists to help developers choose the right NoSQL database for their project that would guarantee high availability and reliability?

RQ2: How does the empirical evidence found, compare with each other in terms of their approach, methods and constraints used in addressing RQ1?

RQ3: What is the strength of the empirical data found for each solution or approach used?

1. What implications would these findings have?

RQ4: What are the implications of the findings for the design of the new empirical data framework of what has been found in RQ3?

I. What is the gap that could be created or exploit to create a solution after looking at other solutions?

A. Defining the systematic literature review protocol

The goal to define the systematic review was to specify systematically how the review process works. To see how each stage of the process unfolds, and to focus on the objective of the review process. Before moving on to the next stage, each stage of the review process was reviewed. The study specified the steps before starting any stage in order to keep the structure and integrity of the review process and the objective of the systematic review.

Search for Relevant Studies

The research process had a goal to investigate the NoSQL data store journal articles to choose which article gets accepted based on the research focus [13].

The main goal of this stage was to investigate and retrieve all the literature that's relevant to answer the research questions. The study then created a search strategy to specify the articles and papers that needed to be searched.

III. SEARCH STRATEGY

The study followed a two-step procedure for the search strategy:

Gather a list that includes sources that would be relevant studies for the review to decide how the study would search these sources.

The study used the university library online research databases to create the list of sources. Added relevant journals searched online manually that could be used for the studies as part of the review. The study made sure that all the sources are active and available by importing them into mendeley reference management software for the citations.

The search process was restricted to search only for published journal articles and papers presented at conferences as they have been validated and published. The query strings applied based on the titles and abstract, papers from 2014-2017 was searched using the online research database. The online research databases used included ACM digital, Emerald insight, IEEE Explore, Science Direct and Springer Link.

The four groups formed the search string these were the key terms used. From the search terms the study created the four groups and added synonyms to search similar words that have the same meaning or related semantic meanings within the study literature. Directly related to the first research question they are uniquely gathering different sets of research studies: Group 1 finds all the studies that focuses on NoSQL data stores; Group 2 finds the literature related to databases; Group 3 find all the studies relating to relational databases; Group 4 finds literature about programming data structures. The study then used a Venn diagram to create an intersection of the four sets. The study used a Boolean expression for the search strategy by using the logical OR-operator within the groups to focus the studies to include any of the terms used per group, then put the groups together with an AND-operator. The study used at least one of each of the terms in each group when performing the search string. Practically this allowed the study to use the one search string to perform 48 different searches as shown below:

$([G1, T1] \text{ OR } [G1, T2]) \text{ OR } ([G1, T3] \text{ AND } ([G2, T1] \text{ OR } [G2, T2] \text{ OR } [G2, T3]) \text{ OR } ([G2, T4] \text{ AND } ([G3, T1] \text{ OR } [G3, T2] \text{ OR } [G3, T3] \text{ OR } [G3, T4] \text{ AND } ([G4, T1] \text{ OR } [G4, T2]))$

TABLE 1

Year	Online research databases / Digital libraries						Total
	IEEE	ACM	Science Direct	Emerald	Springer	Scholar	
2010							
2011	1						
2012							
2013	1			1			
2014	3			1		2	
2015		2			2	1	
2016	3				1		
2017	3		2				
Total	11						
Analyzed	11	2	2	2	3	3	23

II. SELECTION OF STUDIES

The selected research papers based on the online research database search query, articles about NoSQL and relational databases and data structures was selected, the study checked the title that's relevant to the study and the research and filtered the search query. The study ignored books and removed articles that was not published and focused on studying the abstracts and the conclusion of these papers [13].

The goal was to filter down the studies found during the search stage. There were studies relevant to answer the research questions. The study used a set of inclusion criteria and quality screening criteria.

The inclusion criteria used a three stage process:

1. Abstract inclusion criteria screening
2. Full-text inclusion criteria screening
3. Full-text quality screening

The selection process refers to the Primary Inclusion Criteria as the set containing IC1 and IC2, the Secondary Inclusion Criteria contains sets IC3, IC4 and IC5 and the Quality Screening Criteria contains sets QC1 and QC2.

The study filtered studies based on reading the abstracts of the 61 papers found during the search phase from 2010 - 2017, only if the studies in the abstract included the first two inclusion criteria they would have been accepted for the next stage of the study.

IC1 The study's main concern is NoSQL data stores and problems they solve

IC2 The study does present empirical results

Based on the inclusion criteria above only papers from 2014 to 2017 was selected containing the IC1 and IC2, due to time constraints the study had to filter the results by from 2014 till present. The final result of the abstract filtering rejected 29 articles and 23 articles was passed to the full-text screening.

A. Full-text inclusion Criteria Screening

The study filtered out the articles that failed to meet the following from the inclusion criteria:

IC3 The study focuses on choosing the right NoSQL database methods used

IC4 The study focus on comparing NoSQL and RDBMS data structures

IC5 The study looks at challenges NoSQL present

The study couldn't get all the full details of the articles using only the abstracts, therefore needed the full-text inclusion criteria screening, the study applied the same strategy used for the abstract inclusion criteria. The final results of this stage was that the study rejected 8 articles leaving 15 articles for the final stage of the study selection process.

B. Full-text Quality Screening

The final stage of the study selection filtered out studies that did not meet the following quality criteria:

QC1 Is there a clear statement of the aims of the research?

QC2 Is the study put into context of other studies and research?

In this stage the study assessed the remaining articles to see If these articles meet the quality criteria. All the 15 articles passed the quality screening criteria and they all formed the literature basis for the review.

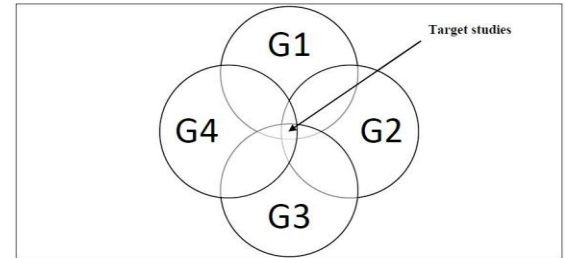


Fig. 1 : Venn Diagram

TABLE 2

	Group 1	Group 2	Group 3	Group 4
Term 1	NoSQL	Databases	Relational	Data structures
Term 2	Not-only SQL	Data stores	RDBMS	hierarchical
Term 3	Non-relational	Databanks	Relative	
Term 4		Records		

TABLE 3

Criterion Identifier	Criterion
IC1	The study's main concern is scaling NoSQL data stores
IC2	The study does present empirical results
IC3	The study focuses on choosing the right NoSQL database
IC4	The study focus on comparing NoSQL and RDBMS data structures
IC5	The study looks at challenges NoSQL present
QC1	Is there a clear statement of the aims of the research
QC2	Is the study put into context of other studies and research

III. QUALITY ASSESSMENT

To focus on the RQ3 the study looked at the strength of evidence presented by the studies in the review, the

study accessed the quality of each study using the following criteria:

- QC1 Is there a clear statement of the aims of the research?
- QC2 Is the study put into context of other studies and research?
- QC3 Are system/algorithm design decisions justified?
- QC4 Is the test data set reproducible?
- QC5 Is the study algorithm reproducible?
- QC6 Is the experimental procedure thoroughly explained and reproducible?
- QC7 Is it clearly stated in the study which other algorithms the study algorithm(s) have been compared with?
- QC8 Are the performance metrics used in the study explained and justified?
- QC9 Are the test results thoroughly analysed?
- QC10 Does the test evidence support the findings presented?

III. RATIONALE FOR THE CRITERIA

The first two criteria were used in the study selection process, these two are considered a good research practice as they clearly identify the aims of the research and allows the current research to be in context of the other research. The third criteria check that the current process or algorithms used are properly analysed, this allows the researcher to analyse an approach before conducting experiments on it. The criteria QC4, QC5, QC6 and QC8 are related to reproducible research, to allow other researchers to reproduce each step done of the work done by the study to compare results to their own approaches. Criteria QC7 looks at the approaches done and how the study compared the results of other studies against the current study. Criteria QC9 and QC10 checks if the results are thoroughly analysed as a quality assessment to check if the findings are supported by the evidence presented in the study. There should be a clear connection between the evidence presented and the conclusions of the study.

IV. DATA COLLECTION

The following data was extracted from the studies included in the review:

1. Name of author(s)
2. Title
3. Study identifier
4. Year of publication
5. Type of article
6. Aims, objectives and contributions of study
7. Experimental design
8. Performance metric used
9. Findings and conclusions

Papers selected for the final study of the research process would allow researchers to separately read the study and fill in the quality assessment form and resolve any disagreements in the quality assessment criteria.

V. DATA ANALYSIS

The study could perform an analysis based on the quality assessment and the data collected. The study then sorted the studies by the solution they solved for the study, next

The study answered the research questions given using the solutions found in the studies and lastly used the quality assessment and data collection tables to evaluate any additional findings that's not related to the research questions.

TABLE 4

Articles	Experiment performed	Bench mark	Client tests	language drivers	Platform used
[15]	yes	YCSB	NO	no	no cluster
[16]	yes	YCSB	AMAZON EC2 INSTANC E	no	amazon ec2 cloud
[1]	yes	YCSB	NO	no	no cluster
[16]	yes	YCSB	AMAZON EC2 INSTANC E	no	amazon ec2 cloud
[17]	yes	YCSB	YES	no	no cluster
[18]	yes	YCSB	NO	no	cluster
[19]	no	NONE	NO	no	review
[20]	yes	YCSB	NO	no	no cluster
[5]	yes	YCSB	NO	no	amazon ec2 cloud

A. Results of the review

Nine articles were selected for the final selection. Upon a closer look, the final selection revealed that eight of the articles all performed experiments. Benchmark tests were performed on eight of the articles selected. None of the articles selected tested the programming language drivers used in NoSQL data stores

To answer the research questions, the study investigated the selected papers in order to provide a broader understanding of the research aim and to highlight the research questions. Researchers have a challenge to determine which data model they should use for applications [15]. The study also explored different workloads studying the latency, throughput and runtime using a Yahoo Cloud Serving Benchmark tests. Modern web applications require high availability and low latency to handle big data sets [16].

The datasets were executed using an experimental research setup on a single workstation. [15] used three different databases, one SQL database and two other NoSQL databases. More studies should be conducted to cover more data models as not all of the NoSQL models were used in this

research. [15], found that SQL is ideal for applications that just read mostly from databases. HBase, which is a NoSQL column-family store database, utilized more CPU resources and writes in memory of average 4GB of RAM. MongoDB, which is a document store NoSQL database, had a high allocation of memory used with an average of 8GB RAM. More studies would need to be done for the rest of the NoSQL data stores and choosing the right database system for an application would still be a high critical task [15].

Some studies analyzed the role of NoSQL and the challenges it presents when having to choose the right NoSQL database. The architecture decision should be made based on the requirements as the features of NoSQL databases are not always the same and can't make any general comparisons [18].

It's impractical to prototype a design for production as this would require hundreds of servers and new databases are constantly emerging [21]. The aim of the research used two-use cases first was to retrieve recent medical results for a particular patient; second was to read a new medical test result from all locations about a medical patient updated. [21] compared the performance and scalability using one server scaling to nine instances. The authors used three NoSQL data stores, MongoDB, which is a document store; Cassandra which is a column store and Riak, which is a key-value store database.

The nine instances replicated a distributed environment across three data centers. In addition to the studies described above, Amazon EC2 cloud instances had to be created to run the database servers and the test client was also executed using an Amazon cloud instance. [21] used the Yahoo Cloud Serving Benchmark for their test client to test latency and throughput. The test client used a selected workload and ran it three times and the client threads were simulated from 1 -10, 25, 50, 100-500, 1000. Using the single node made the research impractical to use for a production environment. Clearly, this showed that they wanted to illustrate which database can distribute the records efficiently and use the single node to give insight why scaling and adding more nodes would be better than using faster nodes with more storage [21].

More studies would need to be done for the rest of the NoSQL data stores and choosing the right database system for an application would still be a high critical task. Some studies analyzed the role of NoSQL and the challenges it presents when having to choose the right NoSQL database. The architecture decision should be made based on the requirements as the features of NoSQL databases are not always the same and can't make any general comparisons [20].

All the 21 papers from 2014 to 2017 that were used in this study, did not focus directly on the programming language drivers that are explored by developers to store, read and write to databases. The papers don't provide a clear and reliable answer to the research question about the best performing language driver to use.

In summary, there is no evidence in the analyzed scientific articles. NoSQL data stores have advantages over relational

databases management systems as they can scale on commodity hardware [5]. Using inexpensive commodity hardware, NoSQL scaling horizontally proves to be cost effective handling large volumes of data, to distribute the data for replication [4]. NoSQL data stores can also be defined as fault-tolerance to handle server failures and still continue uninterrupted, one advantage over relational databases systems is that NoSQL do not have data structure requirements like relational database management systems, they can virtually support any data structure [4].

VI. FINDINGS OF LITERATURE REVIEW

It is clear from the above discussion that while studies have been performed on performance and reliability of distributed systems and commodity hardware, commodity hardware or low-end cost-efficient server hardware are the building blocks for data centers [17].

Most of the performance benchmarks were done on HPC (High End Performance Clusters). This is an expensive approach, as opposed to using inexpensive commodity computers.

There is limited research done on the programming language use on specific data types and empirical evidence for the existence of associations between code quality programming language choice, language properties, and usage domains, could help developers make more informed choices [22]

Factors that determined programming language adoption: the first would be the use of open source libraries that is available for developers to use; this is the most influential factor for choosing a programming language. The second is the availability of existing code as this outweighs intrinsic or build-in functions such as language simplicity, safety and ranking. The third factor would be the developer experience, libraries and legacy code.

The fourth factor is the company size, as employees at large enterprises that value legacy code more than smaller size companies, simplicity, platform constraints, and development speed matter less. Compared to developers overall, those at larger organizations weigh commercial libraries more and open source libraries less (although open source is still weighted more highly). These factors help developers choose which language to adopt.

Based on the finding of the literature review, NoSQL data stores needed to be reliable, and highly available for this research study. In order for an application to be scalable and responsive it need to use the Reactive Manifesto. High availability applications should use the Reactive Manifesto supported by the CQRS approach design. This will allow the research to focus on what NoSQL database to put on the write-side model and what database to put on the read-side of the model.

CONCLUSIONS

So the need to build scalable, responsive systems would be to include a message driven approach. To make sure the system

has elasticity and be resilient. This approach would make the system more flexible and scalable as well as to be a failure-tolerant system. the existing client-server or n-tier application stacks can't take full advantage of the scalable needs of today's cloud environments. Applications adopting the CQRS principle should be divided into a write-model for input and a read-model for output.

Disk volumes have increased making it impossible to keep everything on just one server, as many technologies are different today than what it was years ago. Within two decades a number of database systems evolved, which includes text management, stream processing and data warehousing. These systems have different requirements than the business data processing used for Structured Query Language (SQL).

The Relational Database Management Systems (RDBMS) used by developers to write code using SQL were designed only for business data processing. If a user wants to perform an upgrade and if the single node running the RDBMS needs to be upgraded a user should be able to do this without taking the database down, this is not possible in the current systems. The database has to be copied from one site to another without stopping transactions

To summarize, in order to achieve high availability a system should have a built-in disaster recovery mechanism and RDBMS need to keep multiple replicas consistently seamless on a grid of systems mapped across regions. This would make all resources available during normal operations, and if failures occur this would only affect fewer resources, but this makes a machine sit and idle waiting to take over if the primary system fails.

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